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Topological Anastomotic Registration via Majorana Zero Modes and Galois Prime Moduli Field Theory

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Abstract

Accurate non-rigid registration of microvascular surfboard anastomoses in reconstructive and plastic surgery requires topological robustness against non-isometric deformations, tissue shearing, and local extrema. Here we establish a mathematical registration framework integrating non-Abelian Majorana Zero Modes (MZMs) with finite Galois prime moduli field theory $\mathbb{Z}_p$. By representing surgical excision boundaries as quasiparticle lattices governed by Clifford algebra self-adjoint operators $\gamma_i = \gamma_i^\dagger$, non-rigid displacement vectors are braided in protected topological sub-spaces. Spatial congruences are strictly evaluated over finite prime rings $\mathbb{Z}/p\mathbb{Z}$, bounding Dirichlet deformation residuals and preserving tissue manifold homeomorphisms under extreme intraoperative strain.

Keywords: Majorana Zero Modes, Non-Abelian Anyons, Prime Moduli, Finite Fields, Anastomosis, Topological Registr

1. Majorana Operator Algebra & Non-Abelian Braiding

Let $\{\gamma_1, \gamma_2, \ldots, \gamma_{2N}\}$ be a discrete set of self-conjugate Majorana operators
assigned to the discrete vertices of the surfboard excision boundary, satisfying the canonical Clifford ant

$$\gamma_i^\dagger = \gamma_i, \quad \{\gamma_i, \gamma_j\} = \gamma_i\gamma_j + \gamma_j\gamma_i = 2\delta_{ij}\mathbf{I}$$

Exchanging adjacent boundary keypoints corresponds to the adiabatic non-Abelian braiding unitary oper
$\mathcal{B}_{ij} \in SO(2N)$, operating on the degenerate topological ground-state space:

$$\mathcal{B}_{ij} = \exp\left(\frac{\pi}{4}\gamma_i\gamma_j\right) = \frac{1}{\sqrt{2}}(\mathbf{I} + \gamma_i\gamma_j)$$

The total fermion parity operator $(-1)^F = \prod_{k=1}^{N}(i\gamma_{2k-1}\gamma_{2k})$ establishes a global topological
invariant that prevents self-intersecting boundary inversions during surgical approximation.

2. Galois Prime Moduli Field Equations for Manifold Drift

To eliminate continuous error accumulation, the residual deformation field $u(x,t)$ is projected onto an anatomical sequence of prime ideals $\mathfrak{p}_m \subset \mathcal{O}_K$ within Galois fields $\mathbb{F}_p$:

$$\mathcal{R}_p(\mathbf{x}_i) \equiv \sum_{j=1}^{M} \left(\|\mathbf{x}_i - \mathcal{T}(\mathbf{y}_j)\|^2 + \left(\mathbf{y}_j^T \mathbf{A} \mathbf{x}_i \right) \right) \pmod{p_m}$$

The regularized velocity field update for registration along the surfboard perimeter is governed by the braided prime differential equation:

$$\frac{d\mathbf{x}_i}{dt} = \sum_j \mathcal{B}_{ij}(\mathbf{y}_j - \mathbf{x}_i) \cdot \left(\alpha + \beta \frac{(i \cdot p_m) \pmod{p_{m+1}}}{p_{m+1}} \right)$$

Under the Chinese Remainder Theorem $\mathbb{Z}/N\mathbb{Z} \cong \prod_m \mathbb{Z}/p_m\mathbb{Z}$, the composite registration trajectory uniquely minimizes the global Riemannian distortion energy without stochastic divergence.

$$\Phi_{\text{total}}(\mathbf{X}, \mathbf{Y}) \equiv \bigoplus_{m=1}^{K} \left(\tfrac{1}{2} \text{Tr}\left(\mathbf{X}^T \mathcal{L}_{\text{Majorana}} \mathbf{X} \right) \right) \pmod{p_m}$$

3. Conclusion

Majorana Zero-Mode non-Abelian braiding coupled with prime moduli arithmetic delivers unconditional topological protection and exact manifold registration for delicate vascular micro-anastomoses.

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